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Course: CISC-555

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Type: Final Project

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Intruder Detector

Ideas

The core idea of this project is to develop a real-time intruder detection and alert system using an Internet of Things (IoT) approach. The system leverages a Raspberry Pi as the central controller for handling sensor input, video capture, and alert mechanisms. It is paired with a SwiftUI-based mobile application to provide a user-friendly interface for monitoring and controlling the system remotely. The database is powered by Supabase, offering secure authentication and real-time data handling. This system aims to enhance home or office security through an integrated solution that detects motion, triggers LED indicators, and records intrusions using a camera module.

Abstract

This project presents the development of a smart security system that combines motion detection, live video streaming, and cloud-connected alerts to monitor environments for unauthorized activity. Utilizing a Raspberry Pi with a connected PIR sensor, camera module, and LED, the system detects motion and responds by activating visual indicators and recording footage. A Flask API bridges communication between the Raspberry Pi and a Supabase backend, which stores authentication and real-time sensor data.

The companion iOS application, built with SwiftUI, allows users to view live camera streams, receive motion alerts, and control system settings remotely. The system was built without the use of AI, instead relying on reliable and efficient traditional programming logic to achieve real-time responsiveness and user interaction.

1. Introduction

In an increasingly interconnected world, the integration of the Internet of Things (IoT) into home and business security systems has opened up new possibilities for real-time surveillance, smart control, and automated alerting. This project, titled Intruder Detection and Alert System, presents a comprehensive solution that leverages IoT devices, sensors, camera modules, and modern mobile app frameworks to detect unauthorized movement.

At its core, the system combines a Raspberry Pi running several hardware control modules including a Passive Infrared (PIR) motion sensor, an LED indicator, and a PiCamera with a SwiftUI-based iOS frontend for user interaction. A cloud database powered by Supabase enables user registration, login, and real-time data retrieval. The entire system is designed with modularity, scalability, and user experience in mind, ensuring that it can be customized or extended for various use cases from home monitoring to small office intrusion detection.

2. Objectives

The primary objective of this project is to design and implement a cost-effective, IoT-based intruder detection and alert system that can be monitored and controlled via a mobile device. The specific goals are including below and these objectives were defined to ensure the system is practical for real-world use and can serve as a foundation for further enhancements in the field of smart home security.

- **Motion Detection:** Use a PIR (Passive Infrared) sensor connected to a Raspberry Pi to detect movement in a monitored area. This sensor is a key component in identifying unauthorized entry or suspicious activity and is chosen for its low power consumption and high sensitivity to infrared signals emitted by living beings.
- **Immediate Response:** Upon detecting motion, the system triggers a red LED light to visually indicate intrusion and immediately starts video recording. This real-time response helps to deter intruders and ensures that evidence is captured from the moment of detection, even before the user is notified.
- **Video Recording and Streaming:** Enable real-time video streaming from the Raspberry Pi's camera module and save recordings locally for review. Users can watch the live feed through the mobile application, while the system also saves recordings for later review, offering both real-time awareness and post-incident analysis.
- **Remote Monitoring:** This app displays real-time camera feeds, allows for user login/logout, and presents system status updates. By using a modern mobile interface, the system becomes easily accessible and user-friendly.
- **Secure Authentication:** Integrate user authentication through Supabase to ensure only authorized users can access or control the system. By using Supabase's email/password login, the application upholds basic security standards suitable for home environments.
- **Scalable Architecture:** The system was designed to be easily extendable for future improvements. The modularity of the backend (e.g., Flask), frontend (SwiftUI), and cloud

integration (Supabase) allows additional features like facial recognition, push notifications, or home automation triggers to be implemented with minimal architectural changes.

3. Implementation Details

The implementation of the IoT-based Intruder Detection and Alert System involves several interconnected components, each responsible for handling a specific aspect of the overall functionality. The system follows a modular architecture that integrates hardware interaction on a Raspberry Pi, backend services via Flask, user authentication and event logging using Supabase, and a SwiftUI-based iOS mobile frontend for real-time monitoring.

3.1 Hardware and Raspberry Pi Setup

The core of the system is a Raspberry Pi that interfaces with a PIR motion sensor, a red LED, and a camera module. When motion is detected by the PIR sensor, the Pi activates the red LED and starts recording using the camera module. The system is wired and configured using GPIO pins and managed through a Python script that concurrently handles recording and streaming.

- The motion detection script continuously monitors input from the PIR sensor.
- When motion is detected, the script sends a status update to Supabase and triggers the camera to begin recording.
- Live video streaming is handled through an MJPEG server, allowing real-time monitoring through the mobile app.

3.2 Backend Services with Flask

A lightweight Flask server is hosted on the Raspberry Pi to expose HTTP API endpoints. These APIs serve multiple purposes:

- **Status Endpoint:** Returns the current motion detection state ("motion detected" or "no motion").
- **Camera Stream Endpoint:** Streams live camera footage to the client via MJPEG.
- **Recording Logic:** Combines stream and camera control into a unified handler that ensures recording is not interrupted by live stream access.

This design allows the mobile app to fetch live data, view camera feeds, and remain in sync with real-world events.

3.3 Supabase Integration

Supabase is used as a backend-as-a-service platform to handle and proper policies were configured to allow authenticated users to update and read specific data fields while keeping the system secure.

- **User Authentication:** The mobile app integrates with Supabase's email/password-based authentication system. JWT tokens manage session state and secure API calls.
- **Real-Time Database Logging:** Upon each motion detection event, the Raspberry Pi sends a POST request to update a status table in Supabase. This update is instantly propagated to all connected clients via Supabase's real-time subscription capabilities.

3.4 SwiftUI iOS Mobile Application

The frontend mobile app was developed using SwiftUI and follows the MVC architecture. Key features are including below and this mobile application allows users to actively monitor their premises from anywhere, in real-time, over a secure connection.

- **Login/Logout Screen:** Enables secure sign-in with Supabase credentials.
- **Live Feed View:** Displays real-time video streamed from the Flask backend.
- **Status Monitor:** Subscribes to Supabase and listens for motion detection events, instantly displaying alert banners when activity is detected.
- **Responsive UI:** Designed to be intuitive and minimal, showing critical information such as recording status, motion alerts, and connection indicators.

3.5 Security Considerations

While the system does not currently include AI-driven facial recognition or advanced encryption layers, several security measures were taken:

- Supabase authentication ensures only registered users can access sensitive data.
- The Flask backend only exposes necessary endpoints and runs on a local/private network unless explicitly configured otherwise.
- Recording is stored locally, minimizing external exposure of private footage.

4. Deliverables

The following deliverables were produced as part of the IoT-based Intruder Detection and Alert System project:

4.1 Hardware Setup and Configuration

- A fully assembled and configured Raspberry Pi system, including:
 - PIR motion sensor for intruder detection
 - Red LED indicator for visual feedback
 - Raspberry Pi camera module for video capture and streaming
- Custom wiring and GPIO configuration to integrate hardware with software logic

4.2 Unified Python Application

- A consolidated Python script that:
 - Continuously monitors motion via the sensor
 - Initiates camera recording when motion is detected
 - Streams live camera feed through a Flask MJPEG server
 - Sends motion status updates to Supabase
- The codebase was developed to optimize concurrency and ensure smooth operation of recording and streaming functions

4.3 Flask Backend API

- A lightweight Flask server deployed on the Raspberry Pi with endpoints for:
 - Real-time motion status (/status)
 - Live video feed (/video_feed)
- API integrated with Supabase for status reporting

4.4 Supabase Backend Configuration

- Supabase project configured with:
 - User authentication via email/password
 - REST API and client SDK usage for seamless backend communication
- Policies were set to allow only authenticated users to read/write their own motion detection logs

4.5 SwiftUI iOS Mobile App

- iOS mobile application developed using SwiftUI and structured via the MVC design pattern
- Key functionality includes:
 - Secure user login and logout
 - Real-time display of motion detection alerts using Supabase's real-time capabilities
 - Embedded camera stream view from the Flask MJPEG server
 - Clean and intuitive user interface to support usability and quick information access

4.6 Documentation and Reporting

- This final project report outlining:
 - System concept, objectives, and implementation details
 - Design decisions, architectural overview, and security practices
 - Real-world use cases and limitations
- Additional documentation includes:
 - Supabase setup steps and schema
 - API route definitions and usage examples
 - Setup and deployment instructions for both backend and frontend.

5. Conclusion

The development of the IoT-based Intruder Detection and Alert System successfully demonstrates the integration of hardware sensors, real-time streaming, cloud services, and mobile application development into a cohesive security solution. The system accomplishes its core goals of detecting motion, alerting users remotely, and enabling real-time monitoring through a user-friendly mobile interface. This project showcases a practical application of Internet of Things (IoT) concepts using a Raspberry Pi as the central hub for sensor input and camera output. The use of Supabase as a backend service provided a powerful, scalable way to handle user authentication and data synchronization without the overhead of managing a dedicated server infrastructure. Meanwhile, the SwiftUI iOS app allowed for a responsive and modern mobile experience, ensuring the user remains informed regardless of location.

A notable achievement in this project was the successful unification of video recording and live streaming functions into a single Python script, overcoming limitations of resource contention between processes. In

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addition, setting up real-time communication between the Raspberry Pi and the mobile app through Supabase offered a seamless and efficient channel for updates.

While the current system does not incorporate artificial intelligence, the foundation laid here opens the door for future enhancements such as face recognition, motion classification, or anomaly detection using machine learning. Further improvements may also include integrating alert notifications (e.g., push or SMS), implementing multiple user roles, and supporting cloud video storage.

Overall, this project not only meets its initial objectives but also provides a flexible, extensible framework for developing more advanced home security or monitoring systems in the future.